

# Solving by completing the square

HANDOUT



Interactive version

## GOAL

## In this lesson

**Rewrite any quadratic equation as  $(x + p)^2 = d$  and solve it.**

## BY THE END OF THE LESSON

- ☐ I can rewrite  $x^2 + bx + c$  as  $(x + b/2)^2 + q$ , also when  $b/2$  is a fraction.
- ☐ I can solve a quadratic equation by completing the square, and tell from  $(x + p)^2 = d$  whether it has two roots, one root or none.

## WHAT YOU NEED FROM BEFORE

- ↔ Lesson 1.1 · Quadratic equations: what we can already solve
- ↔ Operations with fractions
- ↔ Completing the square in simple cases

1 class period • about 45 min

## HOOK

## Backwards from (E)

Lesson 1.1 ended with two faces of one equation:

- (E)  $x^2 - 4x - 5 = 0$
- (D)  $(x - 2)^2 = 9$

We went from (D) to (E) by multiplying out. Now go the other way. Move  $-5$  to the right side:  $x^2 - 4x = 5$ .

Level 1 • threshold

Which number must you add to  $x^2 - 4x$  to get  $(x - 2)^2$ ?

This time we knew which square to aim for. How do you find the right square for an equation you have never seen, like  $x^2 + 6x - 7 = 0$ ?

## EXPLAIN

# Completing the square

## A square and a strip

Take  $x^2 + 6x$ . Draw  $x^2$  as a square with side  $x$ , and  $6x$  as a strip that is 6 wide and  $x$  long.

## Split the strip

Cut the strip into two strips, each  $3 \times x$ . Put one beside the square and one below it. The area is still  $x^2 + 6x$ .

The 3 is **half of 6**.

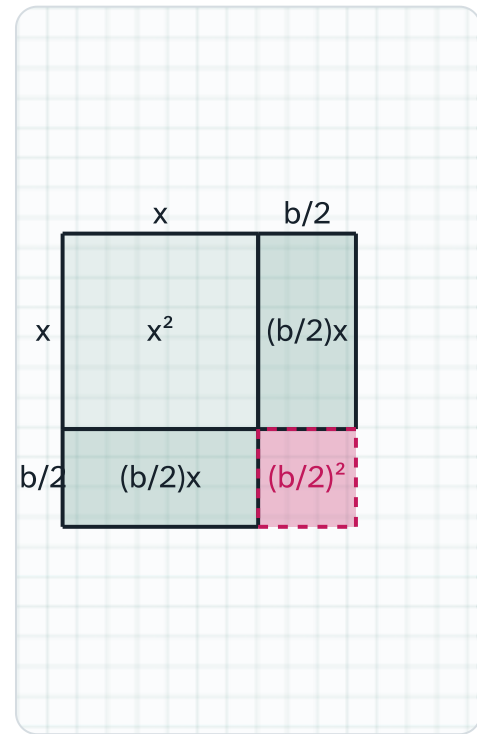
## Fill the corner

One  $3 \times 3$  corner is missing. Adding  $3^2 = 9$  completes a square with side  $x + 3$ :

$$x^2 + 6x + 9 = (x + 3)^2.$$

The drawing only works for  $x > 0$ , but the identity holds for **every**  $x$ : multiply out  $(x + 3)^2$  and you get  $x^2 + 6x + 9$ .

This is called **completing the square**.



## The rule

For any number  $b$ , halve it and square the half:

$$x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(x + \frac{b}{2}\right)^2.$$

Check it by multiplying out the right side:

$$\begin{aligned} \left(x + \frac{b}{2}\right)^2 &= x^2 + 2 \cdot x \cdot \frac{b}{2} + \left(\frac{b}{2}\right)^2 \\ &= x^2 + bx + \left(\frac{b}{2}\right)^2. \end{aligned}$$

- $b$  negative:  $x^2 - 4x + 4 = (x - 2)^2$ . Half of  $-4$  is  $-2$ . The number added is its square,  $(-2)^2 = 4$ , which is positive. (The drawing has no strips of negative width, but the algebra works.)
- $b$  odd:  $x^2 + x + \frac{1}{4} = \left(x + \frac{1}{2}\right)^2$ . Fractions are normal here.

A trinomial with any constant term becomes a square plus a number. Add  $\left(\frac{b}{2}\right)^2$  and take it away again:

$$\begin{aligned} x^2 + 6x + 5 &= \underbrace{x^2 + 6x + 9}_{(x+3)^2} - 9 + 5 \\ &= (x + 3)^2 - 4. \end{aligned}$$

## Solving: four moves

Now solve (E) without knowing (D) in advance. The column on the right explains each step. When we add a number, we add it to **both** sides.

|                    |                                     |
|--------------------|-------------------------------------|
| $x^2 - 4x - 5 = 0$ | $+ 5$                               |
| $x^2 - 4x = 5$     | $+ \left(\frac{-4}{2}\right)^2 = 4$ |
| $x^2 - 4x + 4 = 9$ | left side is a square               |
| $(x - 2)^2 = 9$    |                                     |

Then  $x - 2 = 3$  or  $x - 2 = -3$ , so  $x = 5$  or  $x = -1$ .

**The four moves** for  $ax^2 + bx + c = 0$ :

1. If  $a \neq 1$ , divide both sides by  $a$ . This is allowed because  $a \neq 0$ , and it does not change the roots.
2. Move the constant term to the right side.
3. Add  $\left(\frac{b}{2}\right)^2$  to **both** sides, where  $b$  is now the coefficient of  $x$ .
4. Write the left side as a square,  $(x + p)^2 = d$ , and solve it as in lesson 1.1.

## Two roots, one, or none

Move 4 ends with  $(x + p)^2 = d$ , and  $d$  decides, as in lesson 1.1:

- $d > 0$ .  $x^2 - 2x - 1 = 0$  becomes  $(x - 1)^2 = 2$ . So  $x - 1 = \sqrt{2}$  or  $x - 1 = -\sqrt{2}$ : **two roots**,  $x = 1 + \sqrt{2}$  and  $x = 1 - \sqrt{2}$ . No amount of guessing whole numbers would find them.
- $d = 0$ .  $x^2 - 6x + 9 = 0$  becomes  $(x - 3)^2 = 0$ : **one root**,  $x = 3$ .
- $d < 0$ .  $x^2 + 2x + 5 = 0$  becomes  $(x + 1)^2 = -4$ : **no real roots**.

Two roots are often written together with the sign  $\pm$  ("plus or minus"):  $x = 1 \pm \sqrt{2}$  means  $x = 1 + \sqrt{2}$  or  $x = 1 - \sqrt{2}$ .

## Worked examples

### Example 1 • $x^2 + 6x - 7 = 0$

MOVE C

$$x^2 + 6x = 7.$$

ADD  $(6/2)^2 = 9$  TO BOTH SIDES

$$x^2 + 6x + 9 = 16.$$

WRITE THE SQUARE

$$(x + 3)^2 = 16.$$

SOLVE

$$x + 3 = 4 \text{ or } x + 3 = -4, \text{ so } x = 1 \text{ or } x = -7.$$

CHECK

$$1^2 + 6 \cdot 1 - 7 = 0 \checkmark \text{ and } (-7)^2 + 6 \cdot (-7) - 7 = 49 - 42 - 7 = 0 \checkmark.$$

### Example 2 • An odd b: $x^2 + 5x + 4 = 0$

MOVE C

$$x^2 + 5x = -4.$$

ADD  $(5/2)^2 = 25/4$  TO BOTH SIDES

$$x^2 + 5x + \frac{25}{4} = -4 + \frac{25}{4} = \frac{9}{4}.$$

WRITE THE SQUARE

$$\left(x + \frac{5}{2}\right)^2 = \frac{9}{4}.$$

SOLVE

$$x + \frac{5}{2} = \frac{3}{2} \text{ or } x + \frac{5}{2} = -\frac{3}{2}, \text{ so } x = -1 \text{ or } x = -4.$$

### Example 3 • When $a \neq 1$ : $4x^2 - 8x + 3 = 0$

DIVIDE BY  $A = 4$ 

$$x^2 - 2x + \frac{3}{4} = 0. \text{ The rule needs the coefficient of } x^2 \text{ to be 1.}$$

MOVE C

$$x^2 - 2x = -\frac{3}{4}.$$

ADD  $(-2/2)^2 = 1$  TO BOTH SIDES

$$x^2 - 2x + 1 = \frac{1}{4}, \text{ that is, } (x - 1)^2 = \frac{1}{4}.$$

SOLVE

$$x - 1 = \frac{1}{2} \text{ or } x - 1 = -\frac{1}{2}, \text{ so } x = \frac{3}{2} \text{ or } x = \frac{1}{2}.$$

CHECK

$$4 \cdot \frac{9}{4} - 8 \cdot \frac{3}{2} + 3 = 9 - 12 + 3 = 0 \checkmark \text{ and } 4 \cdot \frac{1}{4} - 8 \cdot \frac{1}{2} + 3 = 1 - 4 + 3 = 0 \checkmark.$$

## SUMMARY

### What to remember

- **Completing the square:** halve the coefficient of  $x$  and square the half:  $x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(x + \frac{b}{2}\right)^2$ .
- Any  $x^2 + bx + c$  is a square plus a number:  $x^2 + 6x + 5 = (x + 3)^2 - 4$ .
- **The four moves** for  $ax^2 + bx + c = 0$ :
  1. if  $a \neq 1$ , divide both sides by  $a$ ;
  2. move the constant term to the right;
  3. add  $\left(\frac{b}{2}\right)^2$  to **both** sides;
  4. write  $(x + p)^2 = d$  and solve it.
- $d > 0$ : two roots,  $x = -p \pm \sqrt{d}$ ;  $d = 0$ : one root,  $x = -p$ ;  $d < 0$ : no real roots.
- Fractions are normal when  $b$  is odd or  $a \neq 1$ . Check the roots by substituting them.

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## NEXT LESSON

**We made the same four moves every time, only with different numbers. What do we get if we make them once, with the letters  $a$ ,  $b$  and  $c$ ?**

[Lesson 1.3 · The quadratic formula →](#)